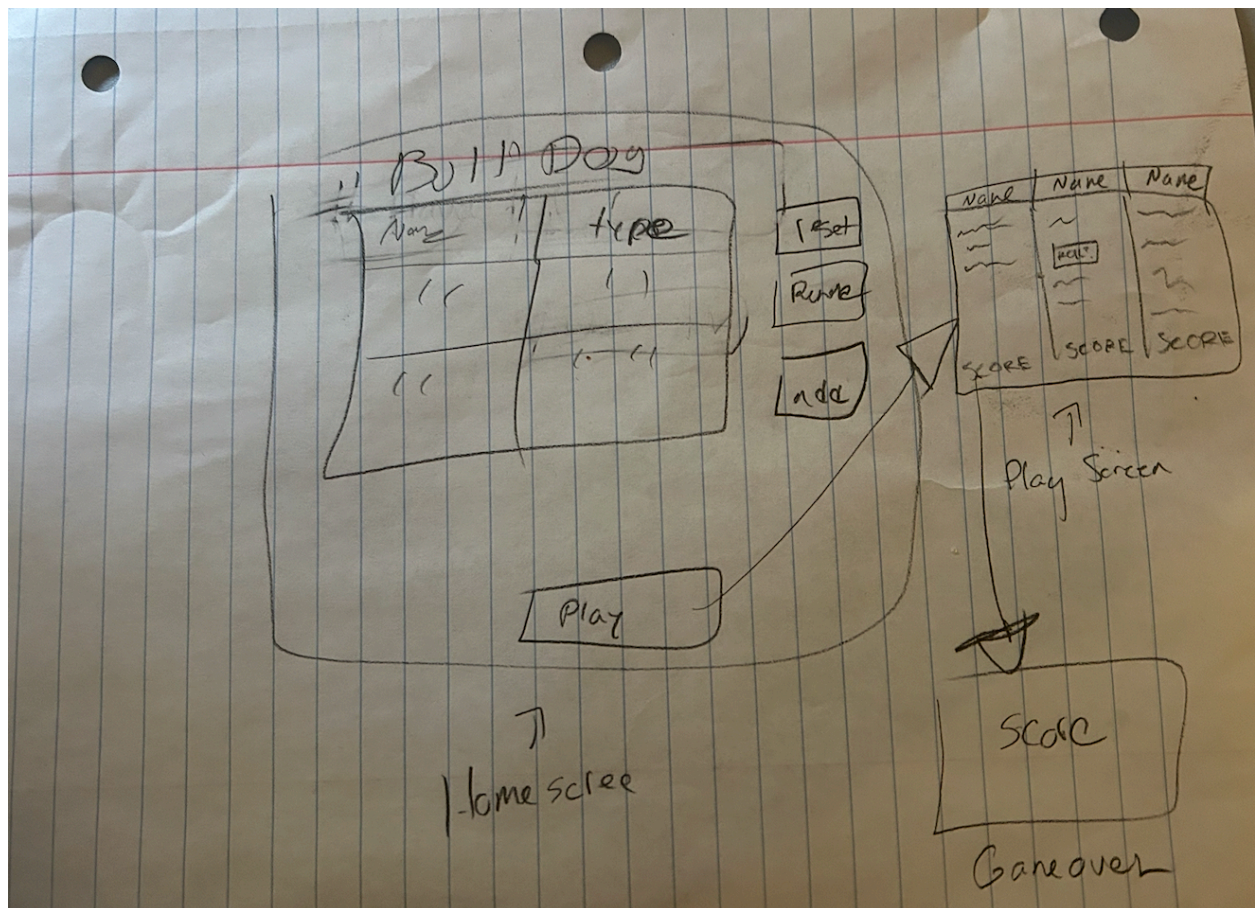


The objective of this assignment was to implement a UI for the bulldog game we have been working on. This paper is divided into three sections; the first section is discussing the design and how it functions. The second section goes over my AI usage. The third section is about my challenges and an overall reflection.

In class we were asked to design a UI for the game with a set of chosen partners. My partners were James Tedder and Nathaniel Serrano. Below I have attached an image of our design.



We choose to break down the UI into 3 screens. The first was the home page. This page includes a title, an interactive table, and four buttons. The interactive table is meant to hold and display the players for the game. It has two columns: One for the player name and one for the type of player. To make the table more functional we added three buttons: add, remove,

and reset. If the add button was clicked it created a new row in the table to change the players name you would double click and type in the players name. For the type if clicked it would open a drop down with the options for the different player types. By default the player type is Wimp, helping ensure that no bug where “no default value” has been. I probably would not have done this if it was not discussed in class on February 19th. The remove button works by selecting the player to be removed and clicking the remove button. The reset button works by clearing the entire table and setting it back to it was at the beginning. The last button is located at the bottom of the UI and is meant to be clicked to start the game.

The second UI screen is the play screen it is meant to display the game as it is being played. It is divided into columns the number of columns corresponding to the number of players in the game. At the top of each column is the player's name. At the bottom of each column it shows the player's current score. In between is an active log of each player's turn.

The third UI screen is the game over screen. It displays the winners name, score, and a button with the ability to play again. I made sure to have this one displayed at the same time that the play screen was active so that way you could go back and see the results of the game.

Below is my AI prompts I used:

02/20/2025 18:42 - Create a Java swing program and make the layout to be a border layout. Have the label be on the top and make it look like it is a title.

I gave it access to home.java

02/20/2025 18:57 - Make a two column table with headers in the center

I gave it access to home.java

02/20/2025 18:58 - Make it so when the add button is selected it will add a player to the table.

I gave it access to home.java

02/20/2025 19:13 - How would I make an option in a table be a drop down menu?

I gave it access to home.java

I personally really liked the AI's response to this question. The idea to use a JComboBox was great.

02/20/2025 19:20 - create java doc

I gave it access to home.java

02/21/2025 15:35 - have it edit the object data each time a round happens and have the score at the bottom update accordingly

I gave it access to playscreen.java

02/21/2025 15:38 - create java doc

I gave it access to playscreen.java

02/21/2025 15:40 - create java doc

I gave it access to gameOver.java

02/21/2025 15:41 - allow it to work properly with human players

I tried to refrain from using AI as much as possible to test my personal skills and give myself a challenge. I currently have a computer science internship that works a lot with UI design. I really enjoy this type of work so I chose to only prompt the AI for the first time I needed to see how something was done. For example when writing the code for the button action listeners. I asked the AI to write me the code for the add button. After seeing the logic of how it did the add button I was able to write the code for the other buttons by myself. However I spent a lot of time trying to figure out the

When it came to testing the program. I tried to code in a way that made it easy for me to test as I went. For example when coding the buttons I made sure to thoroughly test each button before moving on to the next.

What helped me a lot with this assignment was the usage of the Geeks for Geeks website. It was able to offer me a wide range of knowledge on the usage of different Java swing elements.

Overall, this assignment was fairly easy. The way java swing works was pretty intuitive and made logical sense to me. I believe that is due to my knowledge from my internship. Before this assignment I had never used Java Swing before, but I would not be opposed to doing more work with it. I did struggle with the logic for updating the output for the play screen but other than that it was fairly straightforward.

If I were to do this assignment again, I would try to do without the usage of AI. Now that I have a better understanding of Java Swing I think it would provide a good opportunity. I hope to do more work with Java Swing in the future. I feel like I used AI way too much during this assignment and would like to steer further away from it next time.